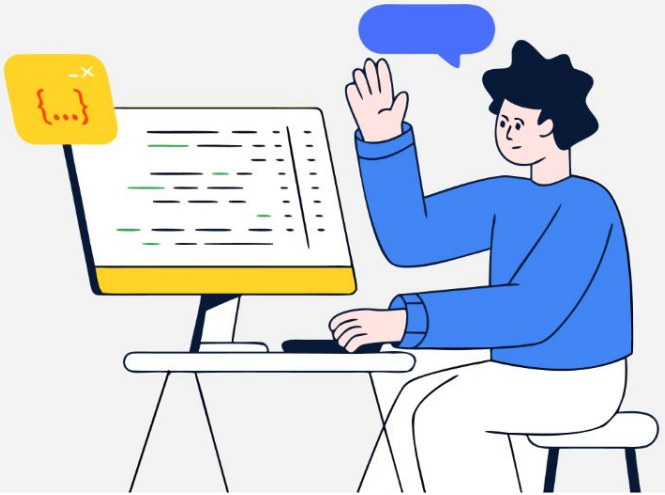
Build with AI

H2S HACK2SKILL

Solution Challenge



Team Details

a. Team name: IDEOVEX
b. Team leader name: Karri Sai Pavani
c. Problem Statement: Zero Touch Rescue AI-based Emergency app

Brief about your solution

ZeroTouch Rescue is a modern AI-powered emergency mobile application designed to provide instant help without requiring manual interaction.
The app focuses on user safety by continuously monitoring real-time signals such as motion, voice, and environmental changes.
It is especially useful in critical situations where the user may not be able to access their phone manually.

- Opportunities
  a. How different is it from any of the other existing ideas?
  b. How will it be able to solve the problem?
  c. USP of the proposed solution.

  Unlike traditional safety apps that rely on manual SOS buttons, ZeroTouchRescue works completely hands-free. Most existing solutions require the user to unlock their phone and trigger an alert, which may not be possible in emergencies.

  The app solves the problem of delayed or unavailable help in critical situations by acting instantly without user interaction.

USP of the proposed solution

🤖 AI-Powered Multi-Sensor Detection (motion, voice, environment)

📡 Works in Low/No Internet (Offline Support)

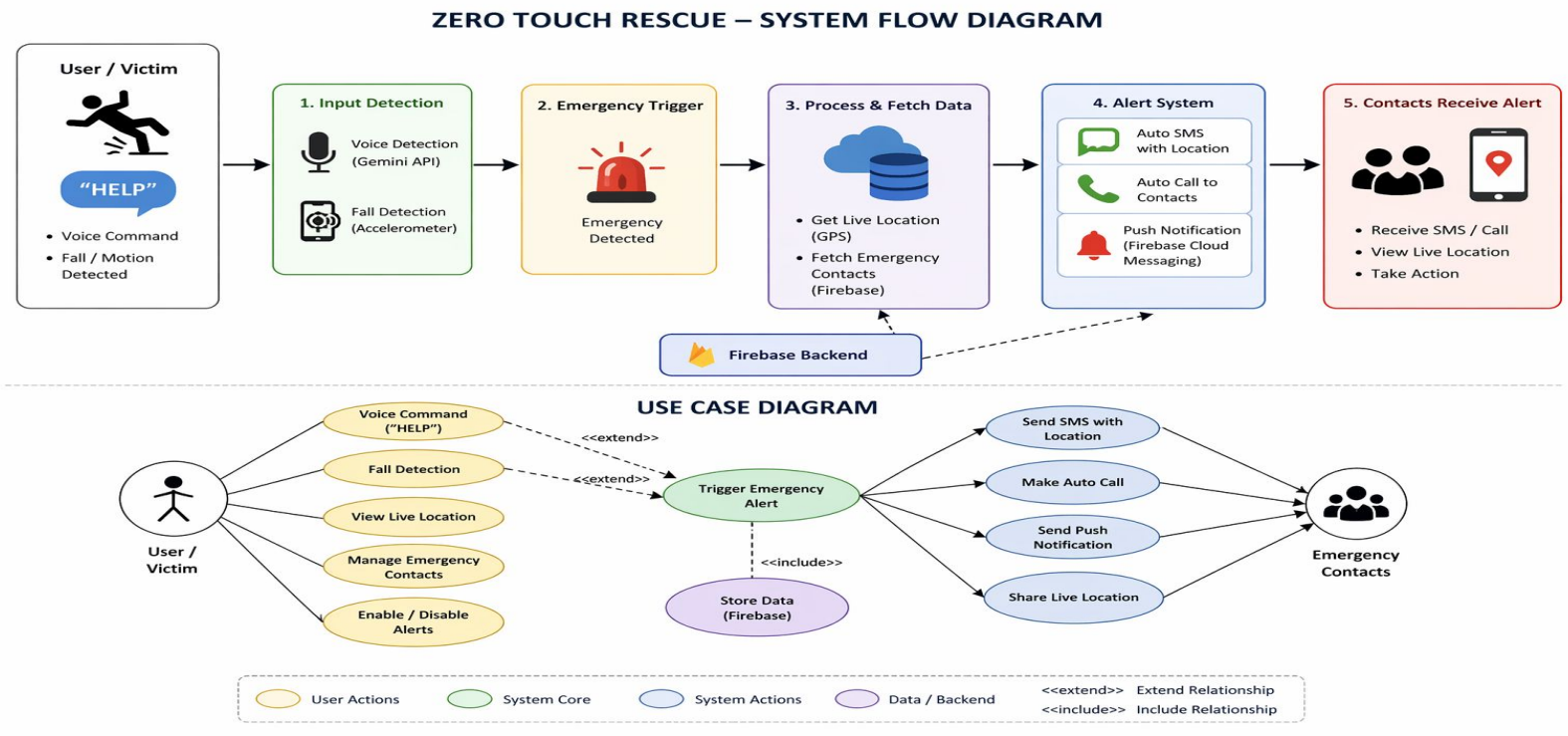
📍 Instant Location Sharing with Alerts

🔒 Secure & Reliable Emergency Communication

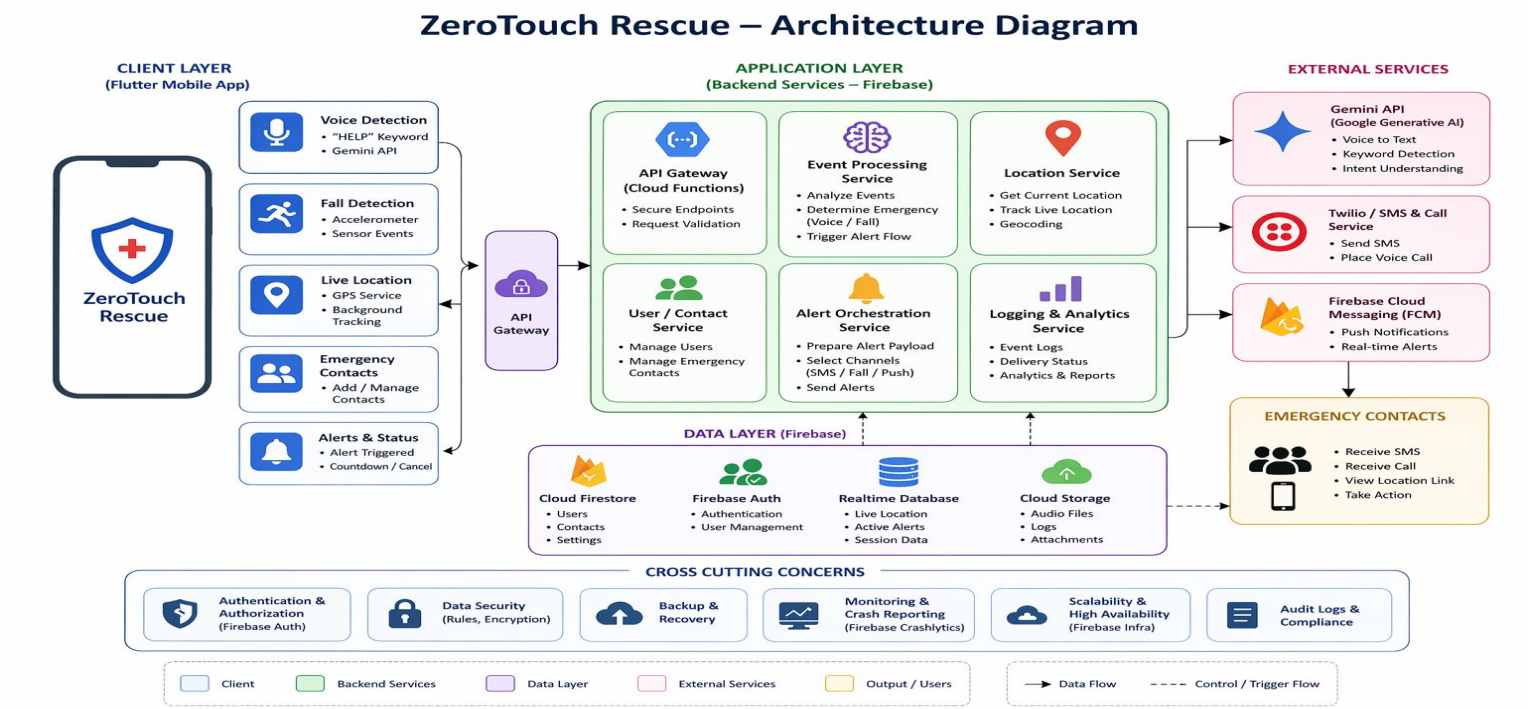
List of features offered by the solution

- Data Security & Privacy
- Emergency Contact Alert System
- Real-Time Location Sharing
- Voice Activation (Panic Detection)
- Offline Mode Support
- Environmental Monitoring
- Smart Safety Check Popup
- Fall Detection System
- Zero-Touch SOS Activation
- AI-Based Emergency Detection

Process flow diagram or Use-case diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution

App Framework: Flutter & Dart (Cross-platform mobile application development)

Artificial Intelligence: Google Gemini AI (Real-time emergency classification & situation analysis)

Backend & Cloud: Firebase Realtime Database (Instant alert synchronization)

Security: AES-256 Encryption (End-to-end protection for sensitive user data)

Hardware Sensors: Accelerometer & Gyroscope (Automated fall and impact detection)

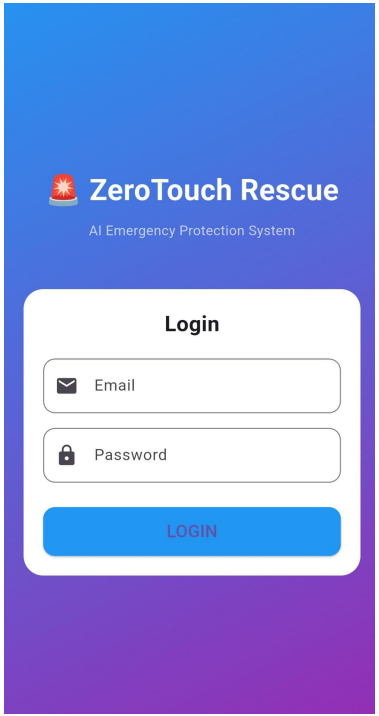
Location Services: Google Geolocator (High-accuracy GPS tracking for rescue)

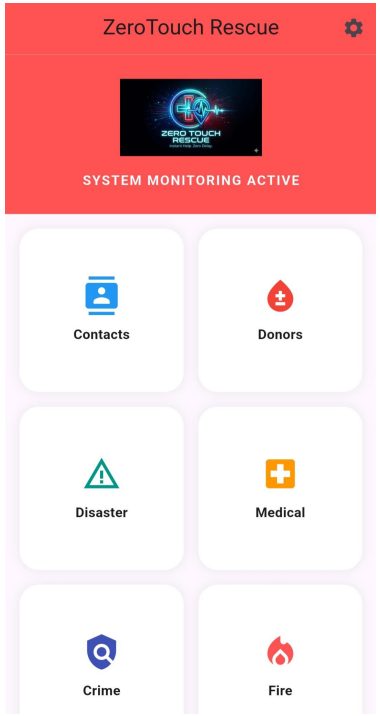
Voice Interface: Speech-to-Text API (Voice-activated emergency triggers: Amma, Kapadandi, Help)

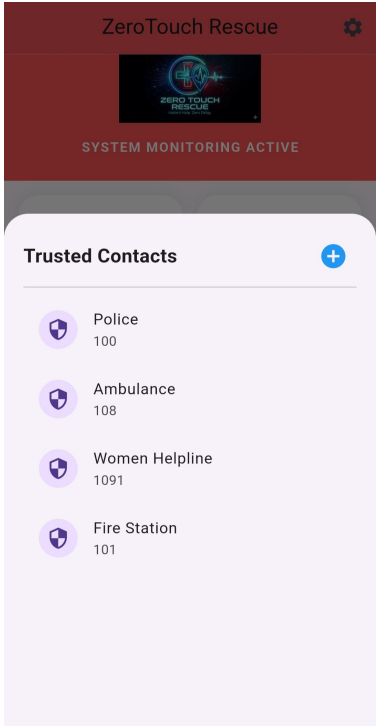
Communication: Fast2SMS API Gateway (Reliable offline SMS fallback for critical alerts)

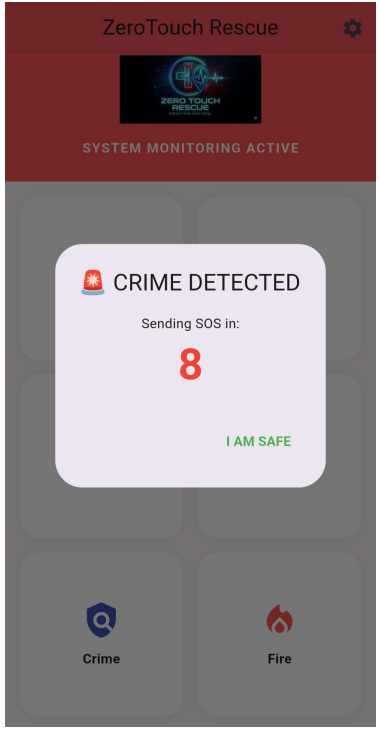
Connectivity: Connectivity Plus (Smart monitoring to switch between Cloud and Offline modes)

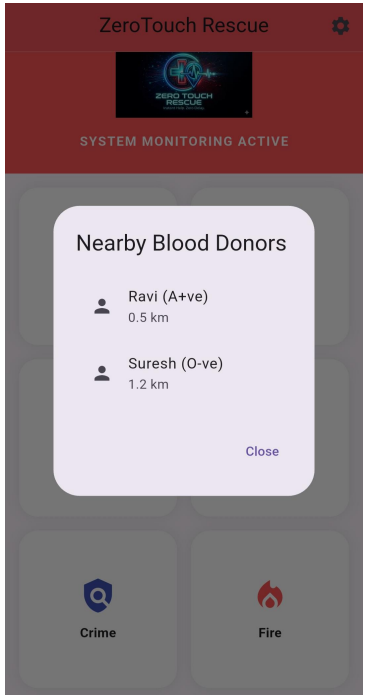
Snapshots of the MVP

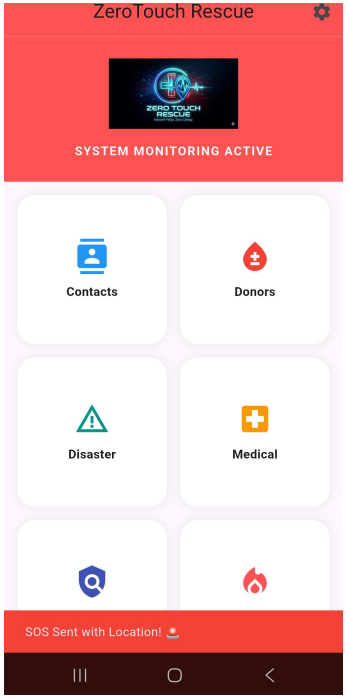


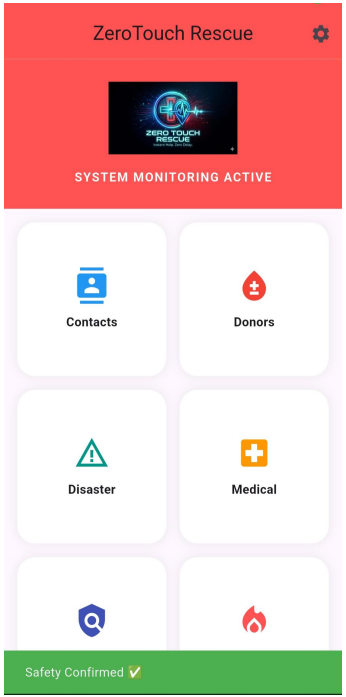


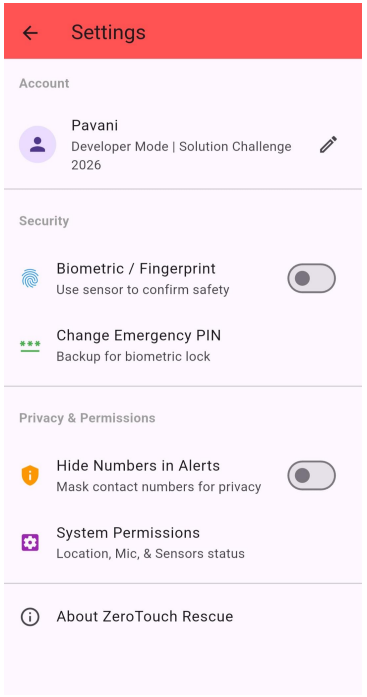





ZeroTouch Rescue
SYSTEM MONITORING ACTIVE
Nearby Blood Donors
Ravi (A+ve)
0.5 km
Suresh (O-ve)
1.2 km
Close
Crime
Fire


ZeroTouch Rescue
SYSTEM MONITORING ACTIVE
Contacts
Donors
Disaster
Medical
SOS Sent with Location!


ZeroTouch Rescue
SYSTEM MONITORING ACTIVE
Contacts
Donors
Disaster
Medical
Safety Confirmed


Settings
Account
Pavani
Developer Mode | Solution Challenge 2026
Security
Biometric / Fingerprint
Use sensor to confirm safety
Change Emergency PIN
Backup for biometric lock
Privacy & Permissions
Hide Numbers in Alerts
Mask contact numbers for privacy
System Permissions
Location, Mic, & Sensors status
About ZeroTouch Rescue

Additional Details/Future Development (if any)

- Multi-Language Voice Detection
- Health Monitoring Integration
- Low Battery Emergency Mode
- Camera-Based Emergency Detection
- Smartwatch / Wearable Trigger
- Background Noise Analysis(present we excluded it due to some permission issues).

Provide links to your:

1. GitHub Public Repository https://github.com/25A31A4282/Zero_Touch_Rescue

2. Demo Video Link (3 Minutes)https://drive.google.com/file/d/1brqWZf6bXmVsxyd6GczHMp4xNYUqLz-X/view?usp=sharing

3. MVP Link https://drive.google.com/file/d/1k_L42SZbJNnUEUbMhAZseej38KUKlm7S/view?usp=sharing

4. Working Prototype Linkhttps://drive.google.com/file/d/1k_L42SZbJNnUEUbMhAZseej38KUKlm7S/view?usp=sharing

Our team members

Karri Sai Pavani(team leader)

Chodisetti Jethendrini Sushma Maheswari

Amara Rashmitha

Kanneedi Sri Samyuktha

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Thank You